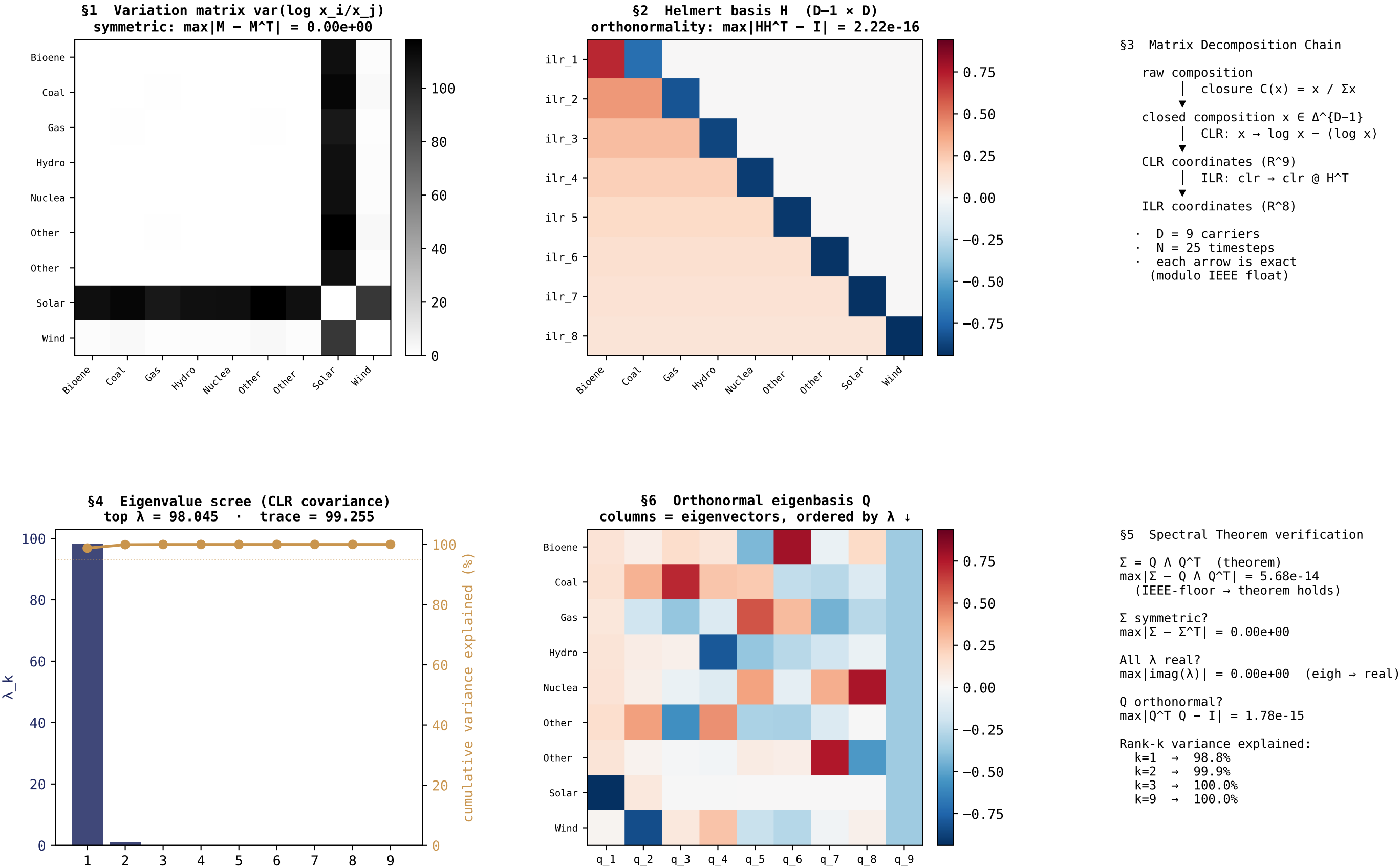


Foundations Plate (Stage 0) – EMB USA United States generation TWh.csv · D = 9 carriers · N = 25 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible



Foundations – Numeric Verification – EMB USA United States generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
\$1 Symmetric	$\max M - M^T $ for variation matrix	0.000e+00	exact (= 0)
\$1 Symmetric	$\max \Sigma - \Sigma^T $ for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
\$2 Transpose	$\max H H^T - I_{\{D-1\}} $ (Helmert orthonormal rows)	2.220e-16	IEEE-floor
\$2 Transpose	$\max H^T H - P_{\{D-1\}} $ (projection onto image)	2.220e-16	IEEE-floor
\$3 Decomp	closure $\rightarrow$ CLR $\rightarrow$ ILR $\rightarrow$ CLR roundtrip residual	2.222e-09	IEEE-floor
\$4 Eigen	number of eigenvalues found	9	= D (9)
\$4 Eigen	$\max \text{imag}(\lambda_k) $ (real because Σ symmetric)	0.000e+00	exact
\$4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	98.0448	–
\$4 Eigen	$\text{trace}(\Sigma) = \sum \lambda_k$ (total variance)	99.2549	–
\$5 Spectral	$\max \Sigma - Q \Lambda Q^T $ (Spectral Theorem residual)	5.684e-14	IEEE-floor
\$5 Spectral	$\max Q^T Q - I $ (orthonormality of eigenbasis)	1.776e-15	IEEE-floor
\$6 Decomp	cumulative variance at k=1 (rank-1 approximation)	98.78%	–
\$6 Decomp	cumulative variance at k=2	99.93%	–
\$6 Decomp	cumulative variance at k=3	99.99%	–
\$7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.